

Qualification Pack



Jr. Technician - Rubberwood Seasoning and Chemical Treatment (Lightwood Furniture)

QP Code: RSC/Q8801

Version: 1.0

NSQF Level: 3

Rubber, Chemical & Petrochemical Skill development Council || 217, 2nd floor, Rectangle One, Saket District Centre



Qualification Pack

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RSC/Q8801: Jr. Technician - Rubberwood Seasoning and Chemical Treatment (Lightwood Furniture)

Brief Job Description

The individual working will be preparing the rubberwood for its use in making lightwood furniture. The person will be responsible for pre-seasoning, chemical treatment, and post-seasoning of rubberwood followed by proper handling and packing process using suitable methods and techniques.

Personal Attributes

This job requires the individual to work independently and be comfortable in performing laborious work. The person should be focused and attentive in performing the assigned task diligently. The individual should be result oriented and positive in attitude. The person must be able to work independently under the guidance of the supervisor.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [RSC/N6204: Perform pre-seasoning of rubberwood](#)
2. [RSC/N4408: Prepare boron solution and perform chemical treatment on rubber wood](#)
3. [RSC/N6205: Carry out post-seasoning of rubberwood](#)
4. [RSC/N6206: Manage the treated sawn timber and pack them safely](#)
5. [RSC/N5626: Follow Environmental and Legal Compliance](#)
6. [RSC/N5007: Follow health & safety protocol](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Natural Rubber (NR) Plantation, Chemical & Petrochemical
Occupation	Chemical Treatment, Production - Rubber Wood, Storage and Handling

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Country	India
NSQF Level	3
Credits	12
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7521.0100, 7521.0200, 7521.0500
Minimum Educational Qualification & Experience	10th grade pass OR 9th grade pass with 1.5 years of experience Relevant OR 8th grade pass with 3 Years of experience Relevant OR 5th grade pass with 6 Years of experience Relevant
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/03/2028
NSQC Approval Date	18/03/2025
Version	1.0
Reference code on NQR	QG-03-RI-03677-2025-V1-RSDC
NQR Version	1

Remarks:

No remarks

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RSC/N6204: Perform pre-seasoning of rubberwood

Description

This unit is about performing pre-seasoning on rubberwood sawn timber to reduce its moisture content so that it is suitable for chemical treatment.

Scope

The scope covers the following :

- Preparing rubberwood sawn timber for pre-seasoning
- Perform pre-seasoning of rubberwood sawn timber

Elements and Performance Criteria

Preparing rubberwood sawn timber for pre-seasoning

To be competent, the user/individual on the job must be able to:

- PC1.** check the quality of sawn and sectioned rubberwood sawn timber
- PC2.** categorize the received rubberwood sawn timber on the basis of its size and its final requirement to make furnitures
- PC3.** measure the moisture content of received rubberwood sawn timber and segregate them accordingly
- PC4.** identify green, partially dried and air dried rubberwood sawn timber
- PC5.** ensure the sorting is done as per the SOP

Perform pre-seasoning of rubberwood sawn timber

To be competent, the user/individual on the job must be able to:

- PC6.** select a suitable process for pre-seasoning
- PC7.** setup the area and equipments as per the method of pre-seasoning chosen
- PC8.** perform pre-seasoning on partially dried and green rubberwood sawn timber using the chosen method
- PC9.** check the moisture content of air dried rubberwood sawn timber
- PC10.** ensure the moisture content of air dried rubberwood sawn timber is in the range of 12-18%
- PC11.** analyse the moisture content of pre-seasoned partially dried and green rubberwood sawn timber
- PC12.** ensure the moisture content of pre-seasoned rubber wood sawn timber is in the range of 12-18%
- PC13.** move the pre-seasoned and air dried rubberwood sawn timber for chemical treatment

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** various grades of rubberwood sawn timber

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- KU2.** about the properties of green rubberwood
- KU3.** about the properties of partially dried rubberwood
- KU4.** about the properties of air dried rubberwood
- KU5.** different types of pre-seasoning methods
- KU6.** the process of air drying the rubberwood
- KU7.** types of kiln drying techniques like dehumidification kiln, solar kiln, vacuum kiln, etc.
- KU8.** the process of kiln drying the rubberwood
- KU9.** setting up procedure for various kilns
- KU10.** about different types of moisture measuring devices
- KU11.** process of measuring moisture of rubberwood using various devices
- KU12.** process of stacking rubberwood sawn timber
- KU13.** how to protect the rubberwood logs from insects, borers and fungi while undergoing preseasoning
- KU14.** the risk and impact of not following defined procedures/ work instructions
- KU15.** the escalation matrix for reporting identified problems
- KU16.** the importance of housekeeping & good shopfloor practices
- KU17.** the health, safety, environment guidelines, legislation and regulations as applicable
- KU18.** the impact of poor practices on health, safety and environment
- KU19.** the potential hazards and actions to minimize the same
- KU20.** the implications of delays in preparation process
- KU21.** the potential problems in preparation process
- KU22.** the indicators and reasons of potential problems
- KU23.** the appropriate solutions to the problems encountered
- KU24.** units of measurement
- KU25.** shelf life of the wood, knowledge of ambient temperature, sunlight and humidity and their effects on wood
- KU26.** the response to emergencies e.g. power failures, fire and system failures
- KU27.** the use of different type of fire extinguishers

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** construct simple sentences and express ideas clearly through written communication
- GS2.** fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS3.** perform functional mathematical operations, including basic mathematical principles, such as numbers and space, and techniques such as: estimation and approximation, for practical purposes
- GS4.** read and understand manuals, health and safety instructions, memos, reports, job cards, images, graphs, diagrams, etc.
- GS5.** understand the various coding systems as per company norms

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- GS6.** communicate with supervisor, upstream and downstream teams, in appropriate manner and using language that is open and respectful
- GS7.** practice honesty with respect to company property and time
- GS8.** resolve any difficulties in relationships with colleagues, or get help from an appropriate person, in a way that preserves goodwill and trust
- GS9.** take responsibility for completing one's own work assignment
- GS10.** take initiative to enhance/ learn skills in ones's area of work
- GS11.** learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one's learning
- GS12.** envisage and articulate personal goals
- GS13.** avoid absenteeism
- GS14.** act objectively, rather than impulsively or emotionally when faced with difficult/ stressful or emotional situations
- GS15.** work in disciplined factory environment and be punctual
- GS16.** take a decision for any change/ issue based on earlier successes (documented previous history) on similar issues
- GS17.** work out changes in case a new improved machine/ equipment is added in the process or any new material is given for replacing the existing one
- GS18.** consult the peer group and superiors to arrive at a favourable decision
- GS19.** use the standard operating procedure or trouble shooting manuals and other reference documents for troubleshooting
- GS20.** review and analyze the process steps to check on system non adherence and non conformity
- GS21.** take a calculated risk with minimum losses
- GS22.** plan work as per the given schedule
- GS23.** complete the assigned task in timely manner
- GS24.** work towards fulfilling the customers requirement as per their demand
- GS25.** ensure timely resolution of the problems, which is possible at his level
- GS26.** communicate effectively to the superior/ customer for any delay in resolving the problem faced by the customer
- GS27.** maintain good/ cordial relation with customers
- GS28.** work on the feedback received from customer regarding the product Interpret quality of product

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preparing rubberwood sawn timber for pre-seasoning</i>	8	22	-	-
PC1. check the quality of sawn and sectioned rubberwood sawn timber	1	3	-	-
PC2. categorize the received rubberwood sawn timber on the basis of its size and its final requirement to make furnitures	2	5	-	-
PC3. measure the moisture content of received rubberwood sawn timber and segregate them accordingly	2	6	-	-
PC4. identify green, partially dried and air dried rubberwood sawn timber	2	5	-	-
PC5. ensure the sorting is done as per the SOP	1	3	-	-
<i>Perform pre-seasoning of rubberwood sawn timber</i>	22	48	-	-
PC6. select a suitable process for pre-seasoning	3	4	-	-
PC7. setup the area and equipments as per the method of pre-seasoning chosen	3	7	-	-
PC8. perform pre-seasoning on partially dried and green rubberwood sawn timber using the chosen method	3	8	-	-
PC9. check the moisture content of air dried rubberwood sawn timber	3	7	-	-
PC10. ensure the moisture content of air dried rubberwood sawn timber is in the range of 12-18%	4	7	-	-
PC11. analyse the moisture content of pre-seasoned partially dried and green rubberwood sawn timber	2	6	-	-
PC12. ensure the moisture content of pre-seasoned rubber wood sawn timber is in the range of 12-18%	2	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. move the pre-seasoned and air dried rubberwood sawn timber for chemical treatment	2	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N6204
NOS Name	Perform pre-seasoning of rubberwood
Sector	Rubber
Sub-Sector	Natural Rubber (NR) Plantation, Non - Tyre
Occupation	Production - Rubber Wood
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	18/03/2025
Next Review Date	18/03/2028
NSQC Clearance Date	18/03/2025

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RSC/N4408: Prepare boron solution and perform chemical treatment on rubber wood

Description

This unit is about preparation of boron solution followed by chemically treating the rubber wood sawn timber using boron solution as well as a suitable technique of impregnation.

Scope

The scope covers the following :

- Preparing boron solution
- Preparing rubber wood sawn timber for chemical treatment
- Performing chemical treatment on rubber wood

Elements and Performance Criteria

Preparing boron solution

To be competent, the user/individual on the job must be able to:

- PC1.** select the process of boron treatment to be opted for chemical treatment of rubber wood
- PC2.** opt out the concentration of boron solution required for chemical treatment of rubber wood as per the process chosen
- PC3.** measure the weight of boric acid and borax accurately as per the required proportion
- PC4.** prepare the boron solution as per the required solution strength
- PC5.** add 0.5% of suitable fungicide (like Sodium Pentachlorophenate) if required
- PC6.** ensure the prepared solution is as per the requirement of the tank
- PC7.** confirm the solution is mixed well and meets the requirements

Preparing rubber wood for chemical treatment

To be competent, the user/individual on the job must be able to:

- PC8.** make sure that the rubber wood are in sizes as per the size of the tank
- PC9.** ensure rubber woods have undergone pre-seasoning
- PC10.** measure the moisture content of the rubber wood using moisture meter
- PC11.** verify that the moisture content of rubber wood picked for chemical treatment is in the range of 12-18%
- PC12.** perform stacking of rubber wood pieces into the fixture if required

Performing chemical treatment on rubber wood

To be competent, the user/individual on the job must be able to:

- PC13.** prepare the equipments required for chemical processing of rubber wood as per the chosen method
- PC14.** operate the equipments as per the standard operating procedure
- PC15.** maintain the process factors like time, pressure, temperature as per the selected method

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- PC16.** make sure the process is completed within the stipulated time and the equipments are shut off
- PC17.** take out the chemically treated rubber wood sawn timber from the tank
- PC18.** determine penetration of chemicals in the rubberwood sawn timber before sending them for post-seasoning

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** various grades of chemical preservatives
- KU2.** selection process of different sizes of rubber wood for chemical treatment
- KU3.** purpose of boron treatment on rubber wood
- KU4.** properties of fungicide like sodium pentachlorophenate
- KU5.** properties of boric acid and borax
- KU6.** different concentrations of boron solutions and their effects on rubber wood
- KU7.** types of impregnation techniques like diffusion treatment, hot and cold open tank method, steam cold quench method, and vacuum pressure impregnation method
- KU8.** effects of various impregnation techniques on rubber wood
- KU9.** time period for various impregnation techniques
- KU10.** factors to be maintained while chemical treatment
- KU11.** cellular structure of rubber wood before and after impregnation process
- KU12.** tools and equipment required for various impregnation methods
- KU13.** effect of chemical treatment on insect and fungal resistance
- KU14.** the risk and impact of not following defined procedures/ work instructions
- KU15.** the escalation matrix for reporting identified problems
- KU16.** the importance of housekeeping & good shopfloor practices
- KU17.** the health, safety, environment guidelines, legislation and regulations as applicable
- KU18.** the impact of poor practices on health, safety and environment
- KU19.** the potential hazards and actions to minimize the same
- KU20.** the escalation matrix and escalation procedure for reporting the hazards
- KU21.** the implications of not adhering the sequence of activities and operations
- KU22.** the implications of delays in preparation process
- KU23.** the potential problems in preparation process
- KU24.** the indicators and reasons of potential problems
- KU25.** the appropriate solutions to the problems encountered
- KU26.** units of measurement
- KU27.** shelf life of the wood, knowledge of ambient temperature, sunlight and humidity and their effects on wood
- KU28.** the response to emergencies e.g. power failures, fire and system failures
- KU29.** the use of different type of fire extinguishers
- KU30.** safety measures to be considered during handling chemicals

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** construct simple sentences and express ideas clearly through written communication
- GS2.** fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS3.** perform functional mathematical operations, including basic mathematical principles, such as numbers and space, and techniques such as: estimation and approximation, for practical purposes
- GS4.** read and understand manuals, health and safety instructions, memos, reports, job cards, images, graphs, diagrams, etc.
- GS5.** understand the various coding systems as per company norms
- GS6.** communicate with supervisor, upstream and downstream teams, in appropriate manner and using language that is open and respectful
- GS7.** practice honesty with respect to company property and time
- GS8.** resolve any difficulties in relationships with colleagues, or get help from an appropriate person, in a way that preserves goodwill and trust
- GS9.** take responsibility for completing one's own work assignment
- GS10.** take initiative to enhance/ learn skills in one's area of work
- GS11.** learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one's learning
- GS12.** envisage and articulate personal goals
- GS13.** avoid absenteeism
- GS14.** act objectively, rather than impulsively or emotionally when faced with difficult/ stressful or emotional situations
- GS15.** work in disciplined factory environment and be punctual
- GS16.** take a decision for any change/ issue based on earlier successes (documented previous history) on similar issues
- GS17.** work out changes in case a new improved machine/ equipment is added in the process or any new material is given for replacing the existing one
- GS18.** consult the peer group and superiors to arrive at a favourable decision
- GS19.** use the standard operating procedure or trouble shooting manuals and other reference documents for troubleshooting
- GS20.** review and analyze the process steps to check on system non adherence and non conformity
- GS21.** take a calculated risk with minimum losses
- GS22.** plan work as per the given schedule
- GS23.** complete the assigned task in timely manner
- GS24.** work towards fulfilling the customers requirement as per their demand
- GS25.** ensure timely resolution of the problems, which is possible at his level
- GS26.** communicate effectively to the superior/ customer for any delay in resolving the problem faced by the customer
- GS27.** maintain good/ cordial relation with customers

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GS28. work on the feedback received from customer regarding the product Interpret quality of product

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preparing boron solution</i>	12	27	-	-
PC1. select the process of boron treatment to be opted for chemical treatment of rubber wood	2	4	-	-
PC2. opt out the concentration of boron solution required for chemical treatment of rubber wood as per the process chosen	2	4	-	-
PC3. measure the weight of boric acid and borax accurately as per the required proportion	3	5	-	-
PC4. prepare the boron solution as per the required solution strength	1	4	-	-
PC5. add 0.5% of suitable fungicide (like Sodium Pentachlorophenate) if required	2	4	-	-
PC6. ensure the prepared solution is as per the requirement of the tank	1	3	-	-
PC7. confirm the solution is mixed well and meets the requirements	1	3	-	-
<i>Preparing rubber wood for chemical treatment</i>	6	17	-	-
PC8. make sure that the rubber wood are in sizes as per the size of the tank	1	3	-	-
PC9. ensure rubber woods have undergone pre-seasoning	1	3	-	-
PC10. measure the moisture content of the rubber wood using moisture meter	2	4	-	-
PC11. verify that the moisture content of rubber wood picked for chemical treatment is in the range of 12-18%	1	5	-	-
PC12. perform stacking of rubber wood pieces into the fixture if required	1	2	-	-
<i>Performing chemical treatment on rubber wood</i>	12	26	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. prepare the equipments required for chemical processing of rubber wood as per the chosen method	2	5	-	-
PC14. operate the equipments as per the standard operating procedure	2	5	-	-
PC15. maintain the process factors like time, pressure, temperature as per the selected method	3	6	-	-
PC16. make sure the process is completed within the stipulated time and the equipments are shut off	2	5	-	-
PC17. take out the chemically treated rubber wood sawn timber from the tank	1	2	-	-
PC18. determine penetration of chemicals in the rubberwood sawn timber before sending them for post-seasoning	2	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N4408
NOS Name	Prepare boron solution and perform chemical treatment on rubber wood
Sector	Rubber
Sub-Sector	Natural Rubber (NR) Plantation, Non - Tyre
Occupation	Chemical Treatment
NSQF Level	3
Credits	4
Version	1.0
Last Reviewed Date	18/03/2025
Next Review Date	18/03/2028
NSQC Clearance Date	18/03/2025

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RSC/N6205: Carry out post-seasoning of rubberwood

Description

This unit is about performing post-seasoning on chemically treated rubberwood to reduce its moisture content so that it can be used to make lightwood furnitures.

Scope

The scope covers the following :

- Verifying and preparing chemically treated rubberwood sawn timber for post seasoning
- Performing post seasoning on rubberwood sawn timber
- Carry out inspection of rubberwood sawn timber post seasoning

Elements and Performance Criteria

Verifying and preparing chemically treated rubberwood sawn timber for post seasoning

To be competent, the user/individual on the job must be able to:

- PC1.** ensure the received rubberwood has been adequately treated with appropriate chemical solution
- PC2.** measure the moisture content inside the chemically treated rubberwood sawn timber
- PC3.** position the rubberwood sawn timber in a well ventilated area away from rain and sunlight
- PC4.** perform stacking of rubberwood sawn timber to facilitate air circulation
- PC5.** check whether the chemically treated rubberwood sawn timber require air seasoning or kiln seasoning on the basis of comparison between current moisture of the treated logs and the EMC (Equilibrium Moisture Content) to be achieved

Perform post-seasoning on rubberwood sawn timber

To be competent, the user/individual on the job must be able to:

- PC6.** keep the rubberwood sawn timbers which needs air seasoning in the ventilated area in stacked position
- PC7.** choose the appropriate kiln seasoning method for post-seasoning of rubberwood
- PC8.** transfer the rubberwood sawn timber which needs kiln seasoning to the kiln method chosen
- PC9.** regulate kiln temperature and humidity to ensure uniform seasoning
- PC10.** increase kiln temperature gradually to prevent cracking/ distortion of rubberwood
- PC11.** maintain appropriate humidity levels during the post-seasoning process
- PC12.** measure the moisture content of the rubberwood during post-seasoning using appropriate moisture meters
- PC13.** ensure the moisture content of the rubberwood reaches in the optimal range of 8-12%
- PC14.** continue kiln seasoning until the required moisture content is achieved
- PC15.** gradually reduce the temperature of the kiln on reaching the required moisture content range allowing rubberwood to cool down evenly
- PC16.** remove the post seasoned rubberwood from the kiln

Carry out inspection of rubberwood sawn timber post-seasoning

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To be competent, the user/individual on the job must be able to:

- PC17.** inspect the post-seasoned rubberwood sawn timber for defects such as warping, cracking or mold growth after successful cooling
- PC18.** measure the moisture content in the post-seasoned rubberwood sawn timber
- PC19.** ensure the moisture content is in the specified range
- PC20.** transfer the rubberwood sawn timber which meets the quality standards to packaging section

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the purpose and methods of chemical treatment for rubberwood, including commonly used chemicals like boron solutions
- KU2.** safety procedures and precautions required when handling chemical treatments
- KU3.** various grades of chemically treated rubberwood sawn timber
- KU4.** different types of post-seasoning methods
- KU5.** the importance of initial air drying to remove surface moisture and how it prepares wood for further processing
- KU6.** principles and mechanics of kiln drying, including temperature and humidity control
- KU7.** types of kiln drying techniques like dehumidification kiln, solar kiln, vacuum kiln, etc.
- KU8.** how to operate and monitor kiln drying equipment effectively
- KU9.** setting up procedure for various kilns
- KU10.** about different types of moisture measuring devices
- KU11.** process of measuring moisture of rubberwood using various devices
- KU12.** process of stacking rubberwood sawn timber
- KU13.** the physical and chemical changes that occur in rubberwood during drying, including the risks of cracking, warping, and distortion
- KU14.** the signs of proper seasoning and potential defects caused by improper drying
- KU15.** the concept of EMC and its importance in achieving long-term stability of the wood
- KU16.** the proper methods for gradually cooling down wood after kiln drying to prevent stress and defects
- KU17.** the criteria for inspecting post-seasoned rubberwood, including common defects and their causes
- KU18.** the standards for acceptable quality and the procedures for handling rejected pieces
- KU19.** the risk and impact of not following defined procedures/ work instructions
- KU20.** the escalation matrix for reporting identified problems
- KU21.** the importance of housekeeping & good shopfloor practices
- KU22.** the health, safety, environment guidelines, legislation and regulations as applicable
- KU23.** the impact of poor practices on health, safety and environment
- KU24.** the potential hazards and actions to minimize the same
- KU25.** the implications of delays in preparation process

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- KU26.** the potential problems in preparation process
- KU27.** the indicators and reasons of potential problems
- KU28.** the appropriate solutions to the problems encountered
- KU29.** units of measurement
- KU30.** shelf life of the wood, knowledge of ambient temperature, sunlight and humidity and their effects on wood
- KU31.** the response to emergencies e.g. power failures, fire and system failures
- KU32.** the use of different type of fire extinguishers

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** construct simple sentences and express ideas clearly through written communication
- GS2.** fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS3.** perform functional mathematical operations, including basic mathematical principles, such as numbers and space, and techniques such as: estimation and approximation, for practical purposes
- GS4.** read and understand manuals, health and safety instructions, memos, reports, job cards, images, graphs, diagrams, etc.
- GS5.** understand the various coding systems as per company norms
- GS6.** communicate with supervisor, upstream and downstream teams, in appropriate manner and using language that is open and respectful
- GS7.** practice honesty with respect to company property and time
- GS8.** resolve any difficulties in relationships with colleagues, or get help from an appropriate person, in a way that preserves goodwill and trust
- GS9.** take responsibility for completing one's own work assignment
- GS10.** take initiative to enhance/ learn skills in one's area of work
- GS11.** learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one's learning
- GS12.** envisage and articulate personal goals
- GS13.** avoid absenteeism
- GS14.** act objectively, rather than impulsively or emotionally when faced with difficult/ stressful or emotional situations
- GS15.** work in disciplined factory environment and be punctual
- GS16.** take a decision for any change/ issue based on earlier successes (documented previous history) on similar issues
- GS17.** work out changes in case a new improved machine/ equipment is added in the process or any new material is given for replacing the existing one
- GS18.** consult the peer group and superiors to arrive at a favourable decision
- GS19.** use the standard operating procedure or trouble shooting manuals and other reference documents for troubleshooting

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- GS20.** review and analyze the process steps to check on system non adherence and non conformity
- GS21.** take a calculated risk with minimum losses
- GS22.** plan work as per the given schedule
- GS23.** complete the assigned task in timely manner
- GS24.** work towards fulfilling the customers requirement as per their demand
- GS25.** ensure timely resolution of the problems, which is possible at his level
- GS26.** communicate effectively to the superior/ customer for any delay in resolving the problem faced by the customer
- GS27.** maintain good/ cordial relation with customers
- GS28.** work on the feedback received from customer regarding the product Interpret quality of product

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Verifying and preparing chemically treated rubberwood sawn timber for post seasoning</i>	12	17	-	-
PC1. ensure the received rubberwood has been adequately treated with appropriate chemical solution	2	2	-	-
PC2. measure the moisture content inside the chemically treated rubberwood sawn timber	3	4	-	-
PC3. position the rubberwood sawn timber in a well ventilated area away from rain and sunlight	2	3	-	-
PC4. perform stacking of rubberwood sawn timber to facilitate air circulation	2	4	-	-
PC5. check whether the chemically treated rubberwood sawn timber require air seasoning or kiln seasoning on the basis of comparison between current moisture of the treated logs and the EMC (Equilibrium Moisture Content) to be achieved	3	4	-	-
<i>Perform post-seasoning on rubberwood sawn timber</i>	24	35	-	-
PC6. keep the rubberwood sawn timbers which needs air seasoning in the ventilated area in stacked position	2	3	-	-
PC7. choose the appropriate kiln seasoning method for post-seasoning of rubberwood	3	4	-	-
PC8. transfer the rubberwood sawn timber which needs kiln seasoning to the kiln method chosen	2	3	-	-
PC9. regulate kiln temperature and humidity to ensure uniform seasoning	3	4	-	-
PC10. increase kiln temperature gradually to prevent cracking/ distortion of rubberwood	2	3	-	-
PC11. maintain appropriate humidity levels during the post-seasoning process	3	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. measure the moisture content of the rubberwood during post-seasoning using appropriate moisture meters	3	4	-	-
PC13. ensure the moisture content of the rubberwood reaches in the optimal range of 8-12%	2	3	-	-
PC14. continue kiln seasoning until the required moisture content is achieved	2	3	-	-
PC15. gradually reduce the temperature of the kiln on reaching the required moisture content range allowing rubberwood to cool down evenly	1	2	-	-
PC16. remove the post seasoned rubberwood from the kiln	1	2	-	-
<i>Carry out inspection of rubberwood sawn timber post-seasoning</i>	4	8	-	-
PC17. inspect the post-seasoned rubberwood sawn timber for defects such as warping, cracking or mold growth after successful cooling	1	2	-	-
PC18. measure the moisture content in the post-seasoned rubberwood sawn timber	1	2	-	-
PC19. ensure the moisture content is in the specified range	1	2	-	-
PC20. transfer the rubberwood sawn timber which meets the quality standards to packaging section	1	2	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N6205
NOS Name	Carry out post-seasoning of rubberwood
Sector	Rubber
Sub-Sector	Natural Rubber (NR) Plantation, Non - Tyre
Occupation	Production - Rubber Wood
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	18/03/2025
Next Review Date	18/03/2028
NSQC Clearance Date	18/03/2025

Qualification Pack

RSC/N6206: Manage the treated sawn timber and pack them safely

Description

This unit is about handling and packing the finally seasoned rubberwood sawn timber with utmost care to prevent it from moisture.

Scope

The scope covers the following :

- Proper handling
- Effective wrapping

Elements and Performance Criteria

Proper handling

To be competent, the user/individual on the job must be able to:

- PC1.** inspect finally seasoned rubber wood sawn timber for any signs of damage, defects, or mold growth before handling and packing
- PC2.** clean the sawn timber to remove surface debris, dust, or residues, etc.
- PC3.** utilize proper equipments like forklifts, pallet jacks, or cranes to handle sawn timber
- PC4.** ensure use of personal protective equipment (PPE), including gloves, safety shoes, and protective clothing
- PC5.** minimize handling frequency to reduce the risk of damage to the sawn timber

Effective wrapping

To be competent, the user/individual on the job must be able to:

- PC6.** stack the rubberwood sawn timber on flat, clean surface without uneven pressure using spacers in between and uniform alignment
- PC7.** cover stacked sawn timber completely with plastic wrap or shrink wrap to protect from moisture and dirt
- PC8.** place desiccant packets between layers for long-term storage to control humidity and prevent mold growth
- PC9.** use strong, adjustable straps or bands to tightly secure the stacked timber, ensuring straps are tight enough to hold sawn timber in place without causing damage
- PC10.** add edge protectors or padding to the corners and edges of the stack to prevent damage from straps and during handling
- PC11.** ensure other factors like proper ventilation, cushioning, clear labels, constant temperature and constant humidity are also taken care of during transit

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** importance of inspecting rubberwood sawn timber for damage, defects, and mold growth before handling
- KU2.** procedures for cleaning sawn timber to remove surface debris, dust, and residues
- KU3.** how to use different types of equipment used for handling sawn timber, such as forklifts, pallet jacks, and cranes
- KU4.** the proper use of personal protective equipment (PPE) such as gloves, safety shoes, and protective clothing
- KU5.** techniques to minimize handling frequency to reduce the risk of damage
- KU6.** the importance of selecting a flat, clean, and dry surface for stacking sawn timber
- KU7.** how to use stickers or spacers between layers to facilitate air circulation and ensure even weight distribution
- KU8.** proper methods for aligning sawn timber uniformly to prevent warping and bending
- KU9.** benefits of using plastic wrap or shrink wrap to protect sawn timber from moisture and dirt
- KU10.** role of desiccant packets in controlling humidity and preventing mold growth during long-term storage
- KU11.** techniques for using strong, adjustable straps or bands to secure stacked sawn timber tightly
- KU12.** importance and process of adding edge protectors or padding to prevent damage during handling and transportation
- KU13.** the key details to include on labels, such as wood type, treatment information, processing date, moisture content, and handling instructions
- KU14.** the importance of providing special handling or storage instructions to ensure proper treatment throughout the supply chain
- KU15.** the benefits of storing sawn timber in a climate-controlled environment to maintain their condition
- KU16.** the best practices for keeping stacks off the ground on pallets or racks to protect from ground moisture and pests
- KU17.** the importance of ensuring good ventilation around stored sawn timber to prevent condensation and mold growth
- KU18.** the procedures for ensuring transport vehicles are clean, dry, and free of contaminants before loading sawn timber
- KU19.** the techniques for securely loading stacks into transport vehicles to prevent shifting during transit
- KU20.** the use of cushioning materials to protect sawn timber from vibrations and impacts during transportation
- KU21.** the indicators and reasons of potential problems
- KU22.** the appropriate solutions to the problems encountered
- KU23.** units of measurement
- KU24.** shelf life of the wood, knowledge of ambient temperature, sunlight and humidity and their effects on wood
- KU25.** the response to emergencies e.g. power failures, fire and system failures
- KU26.** the use of different type of fire extinguishers

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** construct simple sentences and express ideas clearly through written communication
- GS2.** fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS3.** perform functional mathematical operations, including basic mathematical principles, such as numbers and space, and techniques such as: estimation and approximation, for practical purposes
- GS4.** read and understand manuals, health and safety instructions, memos, reports, job cards, images, graphs, diagrams, etc.
- GS5.** understand the various coding systems as per company norms
- GS6.** communicate with supervisor, upstream and downstream teams, in appropriate manner and using language that is open and respectful
- GS7.** practice honesty with respect to company property and time
- GS8.** resolve any difficulties in relationships with colleagues, or get help from an appropriate person, in a way that preserves goodwill and trust
- GS9.** take responsibility for completing one's own work assignment
- GS10.** take initiative to enhance/ learn skills in one's area of work
- GS11.** learn from experience in a range of settings and scenarios and the capacity to reflect on and analyse one's learning
- GS12.** envisage and articulate personal goals
- GS13.** avoid absenteeism
- GS14.** act objectively, rather than impulsively or emotionally when faced with difficult/ stressful or emotional situations
- GS15.** work in disciplined factory environment and be punctual
- GS16.** take a decision for any change/ issue based on earlier successes (documented previous history) on similar issues
- GS17.** work out changes in case a new improved machine/ equipment is added in the process or any new material is given for replacing the existing one
- GS18.** consult the peer group and superiors to arrive at a favourable decision
- GS19.** use the standard operating procedure or trouble shooting manuals and other reference documents for troubleshooting
- GS20.** review and analyze the process steps to check on system non adherence and non conformity
- GS21.** take a calculated risk with minimum losses
- GS22.** plan work as per the given schedule
- GS23.** complete the assigned task in timely manner
- GS24.** work towards fulfilling the customers requirement as per their demand
- GS25.** ensure timely resolution of the problems, which is possible at his level
- GS26.** communicate effectively to the superior/ customer for any delay in resolving the problem faced by the customer
- GS27.** maintain good/ cordial relation with customers
- GS28.** work on the feedback received from customer regarding the product Interpret quality of product

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Proper handling</i>	16	24	-	-
PC1. inspect finally seasoned rubber wood sawn timber for any signs of damage, defects, or mold growth before handling and packing	3	5	-	-
PC2. clean the sawn timber to remove surface debris, dust, or residues, etc.	3	5	-	-
PC3. utilize proper equipments like forklifts, pallet jacks, or cranes to handle sawn timber	3	5	-	-
PC4. ensure use of personal protective equipment (PPE), including gloves, safety shoes, and protective clothing	3	5	-	-
PC5. minimize handling frequency to reduce the risk of damage to the sawn timber	4	4	-	-
<i>Effective wrapping</i>	24	36	-	-
PC6. stack the rubberwood sawn timber on flat, clean surface without uneven pressure using spacers in between and uniform alignment	4	6	-	-
PC7. cover stacked sawn timber completely with plastic wrap or shrink wrap to protect from moisture and dirt	4	6	-	-
PC8. place desiccant packets between layers for long-term storage to control humidity and prevent mold growth	4	6	-	-
PC9. use strong, adjustable straps or bands to tightly secure the stacked timber, ensuring straps are tight enough to hold sawn timber in place without causing damage	4	6	-	-
PC10. add edge protectors or padding to the corners and edges of the stack to prevent damage from straps and during handling	4	6	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure other factors like proper ventilation, cushioning, clear labels, constant temperature and constant humidity are also taken care of during transit	4	6	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N6206
NOS Name	Manage the treated sawn timber and pack them safely
Sector	Rubber
Sub-Sector	Natural Rubber (NR) Plantation, Non - Tyre
Occupation	Production - Rubber Wood, Storage and Handling
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	18/03/2025
Next Review Date	18/03/2028
NSQC Clearance Date	18/03/2025

Qualification Pack

RSC/N5626: Follow Environmental and Legal Compliance

Description

This unit is about various environmental and legal compliances that has to be followed for felling of rubber wood trees followed by sawing, grading, seasoning, and chemical processing.

Scope

The scope covers the following :

- Legal permits, documentation and sustainable practices
- Environmental regulations, health and safety, and regular monitoring

Elements and Performance Criteria

Legal permits, documentation, and sustainable practices

To be competent, the user/individual on the job must be able to:

- PC1.** ensure necessary permits and licenses from relevant authorities has been secured before felling of rubberwood
- PC2.** maintain accurate records of the source of rubberwood, including ownership documents and permissions
- PC3.** ensure compliance with the Indian Forest Act, 1927, and other local forestry laws
- PC4.** follow sustainable harvesting guidelines, including selective logging and replanting to ensure forest regeneration
- PC5.** conduct Environmental Impact Assessments (EIA) to understand and mitigate potential impacts of logging activities
- PC6.** implement measures to reduce environmental impacts based on EIA findings

Environmental regulations, health and safety guidelines, and regular monitoring

To be competent, the user/individual on the job must be able to:

- PC7.** comply with the Air (Prevention and Control of Pollution) Act, 1981, and the Water (Prevention and Control of Pollution) Act, 1974
- PC8.** ensure pollution control equipment are installed to manage emissions and effluents from seasoning and chemical treatment processes
- PC9.** ensure waste management plan for sawdust, wood chips, and chemical waste are implemented
- PC10.** ensure proper disposal of chemical waste according to guidelines set by the Central Pollution Control Board (CPCB)
- PC11.** store chemicals used for wood treatment in compliance with the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016
- PC12.** properly label, handle, and dispose of chemicals to prevent environmental contamination
- PC13.** follow the Factories Act, 1948, for worker safety and health
- PC14.** ensure compliance with the Energy Conservation Act, 2001, to reduce energy consumption
- PC15.** keep detailed records of all processes and compliance measures

Qualification Pack

PC16. submit necessary reports to regulatory bodies, such as the State Pollution Control Board (SPCB) and CPCB

PC17. ensure transparency in reporting environmental and safety performance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Indian Forest Act, 1927, and other local forestry laws governing the cutting of rubberwood
- KU2.** process for obtaining necessary permits and licenses from the State Forest Department or relevant authorities
- KU3.** types of documentation required to prove the legal sourcing of rubberwood, including ownership documents and permissions
- KU4.** sustainable harvesting guidelines, including selective logging and replanting practices
- KU5.** how to conduct an Environmental Impact Assessment (EIA) to evaluate the potential environmental impacts of logging activities
- KU6.** the provisions of the Air (Prevention and Control of Pollution) Act, 1981, and the Water (Prevention and Control of Pollution) Act, 1974
- KU7.** the types of pollution control equipment required to manage emissions and effluents from seasoning and chemical treatment processes
- KU8.** waste management practices for handling sawdust, wood chips, and chemical waste
- KU9.** guidelines set by the Central Pollution Control Board (CPCB) for the disposal of chemical waste
- KU10.** the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016, and their application to chemical storage and handling
- KU11.** the procedures for proper labeling, handling, and disposal procedures for chemicals used in wood treatment to prevent environmental contamination
- KU12.** the Factories Act, 1948, and its provisions related to worker safety and health
- KU13.** the importance of providing personal protective equipment (PPE) and safety training to workers
- KU14.** about energy-efficient kiln drying methods and their benefits for reducing environmental impact
- KU15.** the Energy Conservation Act, 2001, and its relevance to minimizing energy consumption
- KU16.** emission control systems and their role in capturing volatile organic compounds (VOCs) released during drying
- KU17.** the procedures for conducting regular environmental and safety audits
- KU18.** the types of records that need to be maintained for compliance purposes
- KU19.** the reporting requirements to regulatory bodies such as the State Pollution Control Board (SPCB) and the Central Pollution Control Board (CPCB)
- KU20.** the importance of transparency in reporting environmental and safety performance

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** effectively communicate with stakeholders, including regulatory authorities, colleagues, and local communities, regarding compliance requirements and procedures
- GS2.** identify and resolve compliance issues that may arise during felling, cutting, seasoning, and chemical treatment processes, ensuring adherence to environmental and legal standards
- GS3.** pay close attention to detail when obtaining permits, documenting wood sources, and implementing compliance measures, ensuring accuracy and completeness of records
- GS4.** adapt to changing environmental regulations and legal requirements, and to implement new compliance measures as necessary to meet evolving standards
- GS5.** organize and prioritize tasks related to environmental and legal compliance, ensuring that all necessary permits, documentation, and records are obtained and maintained
- GS6.** identify potential environmental and legal compliance issues before they escalate, and taking proactive measures to address them effectively
- GS7.** collaborate effectively with colleagues, contractors, and other stakeholders to ensure that environmental and legal compliance measures are implemented consistently across all stages of rubberwood processing
- GS8.** assess and manage risks associated with felling, cutting, seasoning, and chemical treatment processes, and to implement measures to minimize potential environmental and legal liabilities

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Legal permits, documentation, and sustainable practices</i>	15	20	-	-
PC1. ensure necessary permits and licenses from relevant authorities has been secured before felling of rubberwood	2	2	-	-
PC2. maintain accurate records of the source of rubberwood, including ownership documents and permissions	3	4	-	-
PC3. ensure compliance with the Indian Forest Act, 1927, and other local forestry laws	3	4	-	-
PC4. follow sustainable harvesting guidelines, including selective logging and replanting to ensure forest regeneration	2	4	-	-
PC5. conduct Environmental Impact Assessments (EIA) to understand and mitigate potential impacts of logging activities	2	2	-	-
PC6. implement measures to reduce environmental impacts based on EIA findings	3	4	-	-
<i>Environmental regulations, health and safety guidelines, and regular monitoring</i>	25	40	-	-
PC7. comply with the Air (Prevention and Control of Pollution) Act, 1981, and the Water (Prevention and Control of Pollution) Act, 1974	3	4	-	-
PC8. ensure pollution control equipment are installed to manage emissions and effluents from seasoning and chemical treatment processes	2	4	-	-
PC9. ensure waste management plan for sawdust, wood chips, and chemical waste are implemented	3	4	-	-
PC10. ensure proper disposal of chemical waste according to guidelines set by the Central Pollution Control Board (CPCB)	2	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. store chemicals used for wood treatment in compliance with the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016	3	4	-	-
PC12. properly label, handle, and dispose of chemicals to prevent environmental contamination	2	4	-	-
PC13. follow the Factories Act, 1948, for worker safety and health	2	3	-	-
PC14. ensure compliance with the Energy Conservation Act, 2001, to reduce energy consumption	2	4	-	-
PC15. keep detailed records of all processes and compliance measures	2	3	-	-
PC16. submit necessary reports to regulatory bodies, such as the State Pollution Control Board (SPCB) and CPCB	2	3	-	-
PC17. ensure transparency in reporting environmental and safety performance	2	3	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5626
NOS Name	Follow Environmental and Legal Compliance
Sector	Rubber
Sub-Sector	Natural Rubber (NR) Plantation, Non - Tyre
Occupation	Generic, Storage and Handling
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	18/03/2025
Next Review Date	18/03/2028
NSQC Clearance Date	18/03/2025

Qualification Pack

RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

- PC1.** undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor
- PC2.** work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy
- PC3.** read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc
- PC4.** prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.
- PC5.** use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** dispose off waste safely and correctly in a designated area
- PC7.** risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace
- PC8.** perform work in a manner which minimizes environmental damage
- PC9.** all procedures and work instructions for controlling risk are followed closely
- PC10.** report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC12.** follow emergency procedures as per company standards and workplace requirements

Qualification Pack

- PC13.** use emergency equipment in accordance with manufacturers' specifications and workplace requirements
- PC14.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC15.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC16.** dispose off medical waste in accordance with workplace requirements
- PC17.** report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

- PC18.** comply with general safety procedures
- PC19.** follow standard safety procedures while handling equipment, hazardous material or tool
- PC20.** check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.
- PC21.** ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure
- PC22.** keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

- PC23.** attend fire drills and other safety related workshops organized at the workplace
- PC24.** be aware of first aid, evacuation and emergency procedures
- PC25.** be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

- PC26.** avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC27.** use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)
- PC28.** handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies on incentives, delivery standards, and personnel management.
- KU2.** occupational safety and health policy followed
- KU3.** emergency evacuation procedure
- KU4.** medical policy
- KU5.** company laws and acts
- KU6.** the risks to health and safety and the measures to be taken to control those risks in the area of work

Qualification Pack

- KU7.** workplace procedures and requirements for the handling of workplace injuries/illnesses.
- KU8.** basic emergency first aid procedure
- KU9.** local emergency services
- KU10.** reporting on accidents, incidents and problems to appropriate authorities.
- KU11.** how to use machines as per standard operating procedure
- KU12.** how to maintain work area safe and secure
- KU13.** use of hazardous materials, tools and equipments
- KU14.** emergency evacuation and first aid procedures to be followed
- KU15.** personal hygiene and fitness requirements
- KU16.** general duties under the relevant health and safety legislation
- KU17.** what personal protective equipment and clothing should be worn and how it is cared for
- KU18.** the correct and safe way to use materials and equipment required for work
- KU19.** the importance of good housekeeping in the workplace
- KU20.** safe disposal methods for waste
- KU21.** methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data which are required for record keeping purpose
- GS2.** report problems to the appropriate person in a timely manner
- GS3.** write descriptions and details about incidents in reports
- GS4.** read instruction manuals for hand tools and equipment
- GS5.** read instructions on work orders and procedures
- GS6.** receive instructions and seek advice from superiors
- GS7.** communicate clearly and effectively with others
- GS8.** take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
- GS9.** work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
- GS10.** make changes in cycle time due to improved process.
- GS11.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
- GS12.** consult the peer group and superiors to arrive at a favourable decision.
- GS13.** use of standard available problem solving techniques for decision making
- GS14.** review and analyze the process steps to check on system non adherence andnon conformity
- GS15.** review the current sop and other standards for continuous improvement to facilitate decision making
- GS16.** take a calculated risk with minimum losses
- GS17.** schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

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- GS18.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS19.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS20.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS21.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS22.** work towards fulfilling the customers requirement as per their demand.
- GS23.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS24.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS25.** maintain good/cordial relation with customers.
- GS26.** work on the feedback received from customer regarding the product.
- GS27.** use first aid treatment in case of any injury/accident.
- GS28.** monitor and maintain the condition of tools and equipment
- GS29.** assess situation & identify appropriate control measures
- GS30.** act, communicate and report in emergency situation

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain a clean and efficient workplace</i>	9	21	-	-
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
<i>Render appropriate emergency procedures</i>	9	18	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
<i>Maintain standard safety procedures at the workplace</i>	6	15	-	-
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
<i>Participate in safety awareness campaigns</i>	2	8	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
<i>Understand potential sources of accidents</i>	1	3	-	-
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
<i>Use safety gears to avoid accidents</i>	3	5	-	-
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N6204.Perform pre-seasoning of rubberwood	30	70	0	0	100	20
RSC/N4408.Prepare boron solution and perform chemical treatment on rubber wood	30	70	0	0	100	30
RSC/N6205.Carry out post-seasoning of rubberwood	40	60	0	0	100	20
RSC/N6206.Manage the treated sawn timber and pack them safely	40	60	0	0	100	10
RSC/N5626.Follow Environmental and Legal Compliance	40	60	-	-	100	10
RSC/N5007.Follow health & safety protocol	30	70	-	-	100	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	230	420	-	-	650	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.